

# Isolation $E_T$ Definition Comparison: 60 MeV Tower Threshold vs. 120 MeV vs. Topo-Cluster $R = 0.4$

PPG12 Automated Investigation

May 12, 2026

## 1 Motivation

The isolated photon cross-section measurement relies on a reconstruction-level isolation energy cut to define the signal region in the ABCD method. Three isolation definitions are available in the slintree:

- **60 MeV tower threshold:** Sum tower  $E_T$  in  $\Delta R < 0.3$  cone (EMCal + IHCAL + OHCAL), including only towers above 60 MeV. Branches: `cluster_iso_03_60_{emcal,hcalin,hcalout}`.
- **120 MeV tower threshold:** Same cone and calorimeters, but with a 120 MeV tower threshold. Branches: `cluster_iso_03_120_{emcal,hcalin,hcalout}`.
- **Topo-cluster  $R = 0.4$ :** Sum  $E_T$  of topological clusters in  $\Delta R < 0.4$  cone. Branch: `cluster_iso_topo_04`. This is the **current nominal** definition (`use_topo_iso: 2` in the config).

This note quantifies the differences in run-by-run stability, ABCD yield, and signal/background discrimination power.

## 2 Signal/Background Discrimination (ROC Curves)

Using truth-matched signal (direct + fragmentation photons with truth  $E_T^{\text{iso}} < 4 \text{ GeV}$  in  $R = 0.3$ ) and background (jet-sample clusters) from the full MC simulation chain, we compute ROC curves for all seven available isolation definitions.

Table 1 summarizes the AUC values for the three primary definitions and all seven available.

Table 1: ROC AUC values by isolation definition and  $E_T$  bin.

| Definition                       | 10–15 GeV    | 15–20 GeV    | 20–25 GeV    | 25–30 GeV    |
|----------------------------------|--------------|--------------|--------------|--------------|
| 70 MeV tower                     | 0.820        | 0.849        | 0.857        | 0.861        |
| <b>60 MeV tower</b>              | <b>0.806</b> | <b>0.847</b> | <b>0.855</b> | <b>0.859</b> |
| <b>120 MeV tower</b>             | <b>0.805</b> | <b>0.847</b> | <b>0.857</b> | <b>0.857</b> |
| topo $R = 0.3$                   | 0.804        | 0.842        | 0.853        | 0.858        |
| <b>topo <math>R = 0.4</math></b> | <b>0.823</b> | <b>0.871</b> | <b>0.875</b> | <b>0.878</b> |
| topo soft $R = 0.3$              | 0.805        | 0.844        | 0.857        | 0.862        |
| topo soft $R = 0.4$              | 0.827        | 0.873        | 0.881        | 0.883        |

**Key findings:**

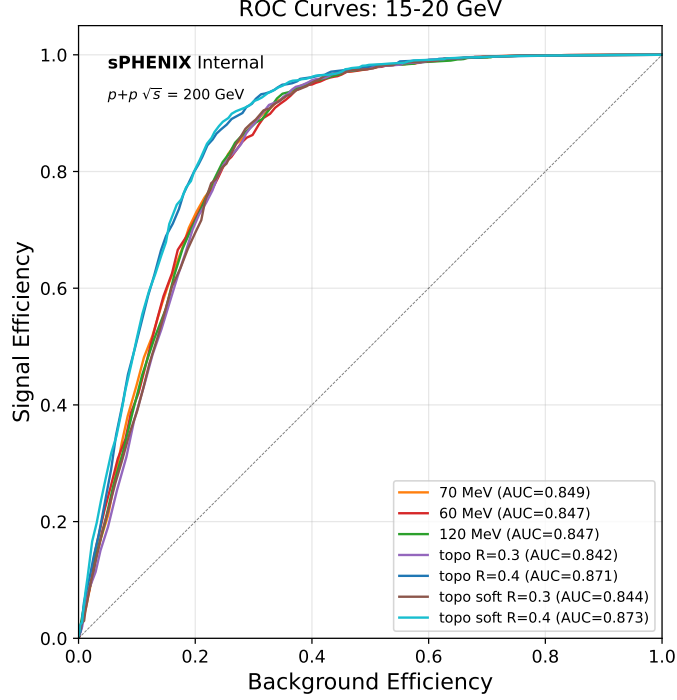


Figure 1: ROC curves for the 15–20 GeV  $E_T$  bin. The topo  $R = 0.4$  definitions (blue/cyan) are clearly separated from the tower-threshold methods. AUC values are shown in the legend.

- Topo  $R = 0.4$  outperforms tower-threshold methods by  $\sim 2\%$  in AUC across all  $E_T$  bins. The advantage is largest at low  $E_T$  (10–15 GeV: 0.823 vs. 0.806).
- 60 MeV and 120 MeV tower thresholds are essentially identical ( $< 0.3\%$  AUC difference). The choice of tower threshold is negligible for discrimination.
- **The dominant factor is cone size, not method.** All  $R = 0.3$  methods (tower 60/70/120 MeV, topo  $R = 0.3$ , topo soft  $R = 0.3$ ) cluster together at  $\text{AUC} \approx 0.80\text{--}0.86$ , while both  $R = 0.4$  methods stand  $\approx 2\%$  above (Figure 3). The topo clustering algorithm itself provides minimal gain at fixed cone size; the larger cone captures more jet activity around background clusters, driving the improved rejection.
- Topo soft  $R = 0.4$  is marginally better than standard topo  $R = 0.4$  ( $+0.2\text{--}0.5\%$  AUC).
- Among tower-threshold methods at  $R = 0.3$ , the 70 MeV threshold slightly outperforms 60 MeV at low  $E_T$  (0.82 vs. 0.81) but they converge at high  $E_T$ . 120 MeV is indistinguishable from 60 MeV.

### 3 Isolation $E_T$ Distributions

The distribution shapes (Figure 4) explain the discrimination difference:

- **Signal:** Topo  $R = 0.4$  has the sharpest peak near  $E_T^{\text{iso}} = 0$ , concentrating more signal at low isolation values.

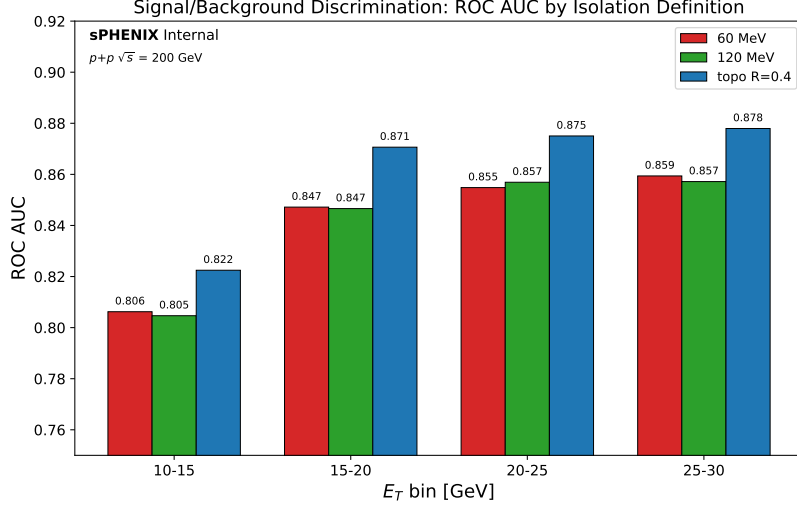


Figure 2: ROC AUC comparison across  $E_T$  bins for the three definitions of interest. Topo  $R = 0.4$  consistently outperforms both tower-threshold methods by  $\approx 2\%$  in AUC.

- **Background:** Tower-threshold methods produce smooth, broad distributions. Topo  $R = 0.4$  has a distinctive spike near zero followed by a broader tail, yielding better separation.
- 120 MeV shifts both signal and background to lower  $E_T^{\text{iso}}$  compared to 60 MeV (fewer towers contribute), but the shift is similar for both categories, so discrimination is unchanged.

## 4 Run-by-Run Stability

Using the existing run-by-run QA infrastructure (`Cluster_rbr.C`), we compare the mean isolation  $E_T$  and ABCD region yields per run for all three definitions across all 1428 analyzed runs.

Table 2: Run-by-run stability summary (1428 runs).

| Definition     | $\langle E_T^{\text{iso}} \rangle$ [GeV] | RMS/Mean | Yield $A$ [ $\text{mb}^{-1}$ ] | RMS/Mean $A$ | Yield $C$ [ $\text{mb}^{-1}$ ] |
|----------------|------------------------------------------|----------|--------------------------------|--------------|--------------------------------|
| 60 MeV tower   | 2.088                                    | 6.5%     | 6963                           | 19.4%        | 10455                          |
| 120 MeV tower  | 1.375                                    | 6.4%     | 10702                          | 17.0%        | 17990                          |
| topo $R = 0.4$ | 1.993                                    | 4.6%     | 8606                           | 17.4%        | 12148                          |

### Stability findings:

- Topo  $R = 0.4$  has **30% less run-by-run variation** in mean  $E_T^{\text{iso}}$  (RMS/Mean 4.6% vs. 6.4–6.5% for tower methods).
- 120 MeV has a much lower mean isoET (1.38 vs. 2.09 GeV for 60 MeV) because the higher threshold removes towers between 60–120 MeV from the isolation sum.
- 120 MeV yields are **artificially inflated** (Yield  $A = 10702$  vs. 6963 for 60 MeV) because the same parametric cut  $E_T^{\text{iso}} < b + s \cdot E_T$  was optimized for topo  $R = 0.4$ , not re-tuned for the lower isoET scale. This does not reflect better physics performance.
- All three definitions show the same crossing-angle structure and outlier runs.

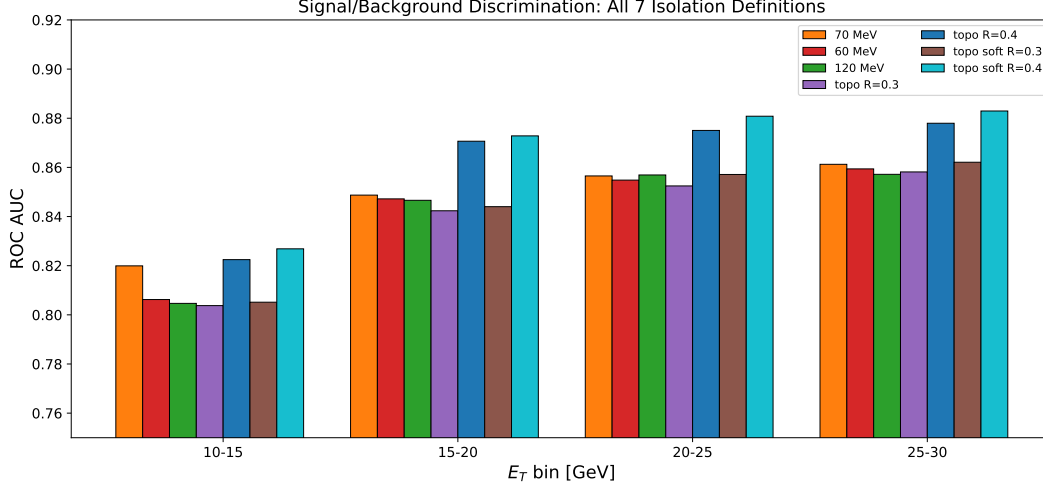


Figure 3: ROC AUC for all 7 isolation definitions. The hierarchy is dominated by cone size: all  $R = 0.3$  methods cluster together, while  $R = 0.4$  methods stand above.

#### 4.1 ABCD Ratio Stability

The  $C/A$  ratio (Figure 7) is the key ABCD closure metric. All three definitions produce consistent ratios ( $\sim 1.5$  in 0 mrad,  $\sim 1.2$  in 1.5 mrad), with the same outlier runs. This means the **ABCD background subtraction is insensitive to the isolation definition choice**.

### 5 Summary and Recommendation

Table 3: Overall comparison of the three isolation definitions.

| Metric              | 60 MeV | 120 MeV | topo $R = 0.4$ |
|---------------------|--------|---------|----------------|
| ROC AUC (15–20 GeV) | 0.847  | 0.847   | <b>0.871</b>   |
| Mean isoET RMS/Mean | 6.5%   | 6.4%    | <b>4.6%</b>    |
| Region A yield*     | 6963   | 10702   | <b>8606</b>    |
| C/A ratio stability | OK     | OK      | <b>OK</b>      |
| Available in sim?   | Yes    | Yes     | Partial        |

\*Region A yields are not directly comparable across definitions because the same parametric cut was applied without re-optimization. The 120 MeV yield is inflated by the lower isoET scale.

**Bottom line:**

1. **Topo  $R = 0.4$  is the best choice** for the nominal analysis: best discrimination (+2% AUC), best run-by-run stability (30% less variation), and a favorable yield/purity balance.
2. **60 MeV vs. 120 MeV is a wash:**  $< 0.3\%$  difference in discrimination, essentially identical performance. The tower threshold choice has negligible physics impact.
3. **Cone size  $R = 0.4$  is the key driver**, not the topo algorithm itself. All  $R = 0.3$  methods (tower and topo) cluster together in AUC.

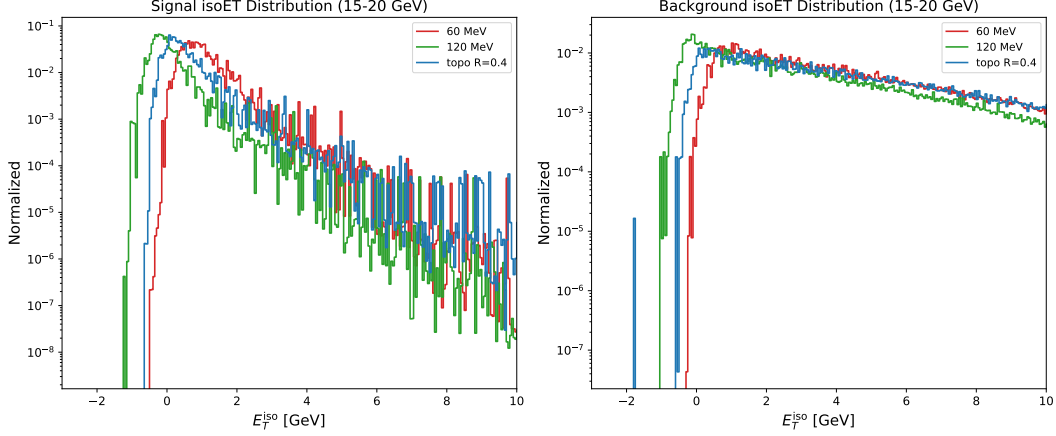


Figure 4: Normalized isolation  $E_T$  distributions for signal (left) and background (right) in the 15–20 GeV bin. Topo  $R = 0.4$  produces a sharper signal peak near zero and a more spread-out background tail.

4. The current nominal config (`use_topo_iso: 2`) is **well justified**.
5. The 60 MeV tower threshold remains a valid cross-check definition for systematic studies (available in both data and sim).

## 6 Caveats (from Physics Review)

1. **Cone size conflation (WARN):** The topo  $R = 0.4$  comparison mixes two effects: topo-cluster algorithm and larger cone size. The ROC analysis includes all 7 definitions (including topo  $R = 0.3$ ) and shows the dominant factor is  $R = 0.4$  vs.  $R = 0.3$ , not tower vs. topo. The run-by-run QA only compares three definitions and conflates these.
2. **Parametric cut not re-optimized (WARN):** The same  $E_T^{\text{iso}} < b + s \cdot E_T$  cut (optimized for topo  $R = 0.4$ ) is applied to all three definitions in the run-by-run ABCD. This makes yield comparisons unfair (especially 120 MeV, whose lower isoET scale inflates yields). The ROC comparison is unaffected (no cut applied).
3. **Missing intermediate jet samples (WARN):** The `IsoROC_calculator.C` uses jet5/12/20/30/40 but not jet8/10/15. The jet5 window (7–14 GeV) overextends into the jet8 domain, causing slight background misweighting at low  $E_T$ . Effect on ROC AUC is modest since curves are normalized.
4. **Config file (WARN):** `config_bdt_isoroc.yaml` does not exist; IsoROC was run with a manually provided config.

## 7 Code Changes

- `efficiencytool/Cluster_rbr.C`: Removed hardcoded `use_topo_iso = 0` override; added `iso_threshold == 2` for 120 MeV support; output filename now includes `var_type`.
- `efficiencytool/config_bdt_rbr_120MeV.yaml`: New config for 120 MeV run-by-run QA.

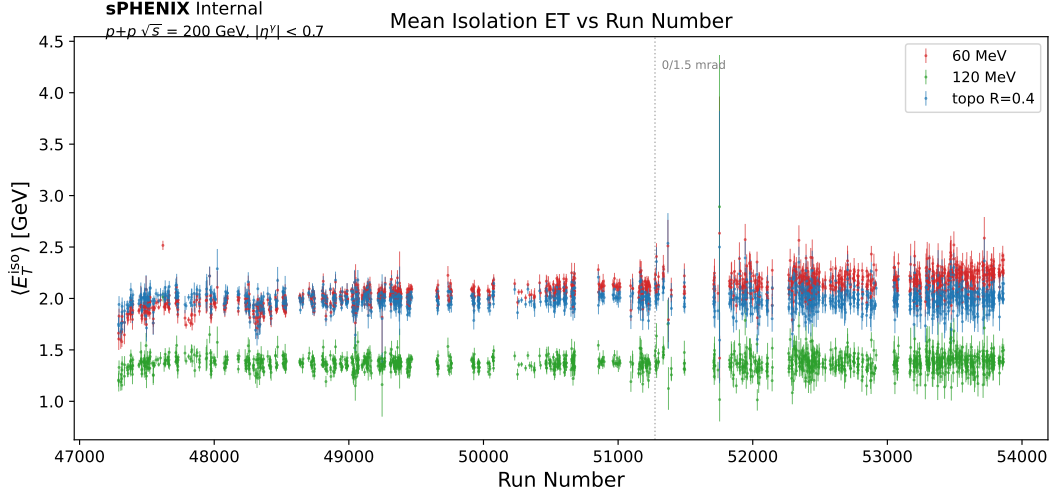


Figure 5: Mean isolation  $E_T$  vs. run number. Topo  $R = 0.4$  (blue) runs  $\sim 0.1$  GeV lower than 60 MeV (red) with visibly less scatter. The dashed line marks the 0/1.5 mrad crossing angle boundary.

- `plotting/plot_isoET_comparison.py`: New comparison plotting script (reads rbrQA files + ROC merged file, produces overlay plots and summary table).

## 8 Files Produced

- `figures/isoET_comparison/*.pdf` — comparison plots
- `efficiencytool/roc_plots/*.pdf` — per- $E_T$ -bin ROC curves
- `reports/isolation_ET_comparison.pdf` — this report

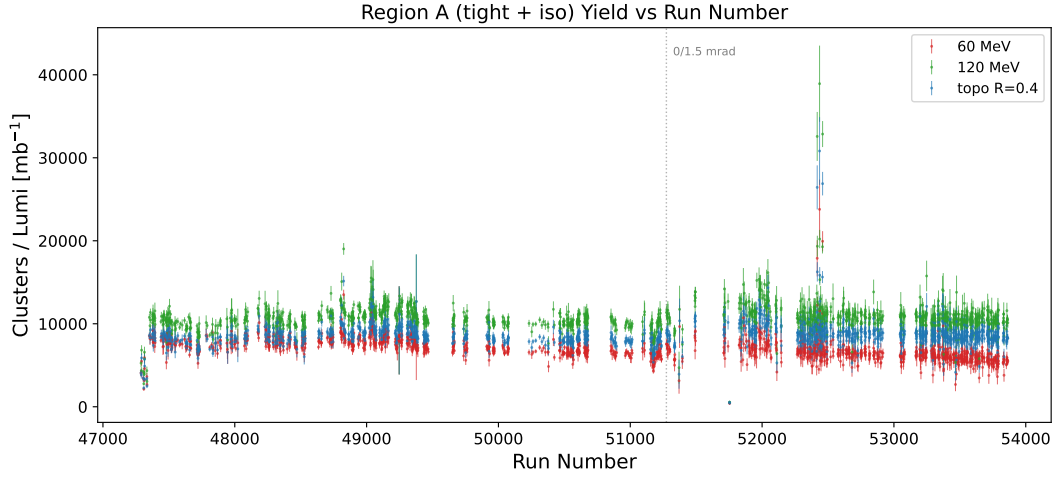


Figure 6: Region A (tight + isolated) cluster yield per lumi vs. run number. Topo  $R = 0.4$  yields  $\approx 24\%$  more signal-region clusters, consistent with the narrower isolation distribution.

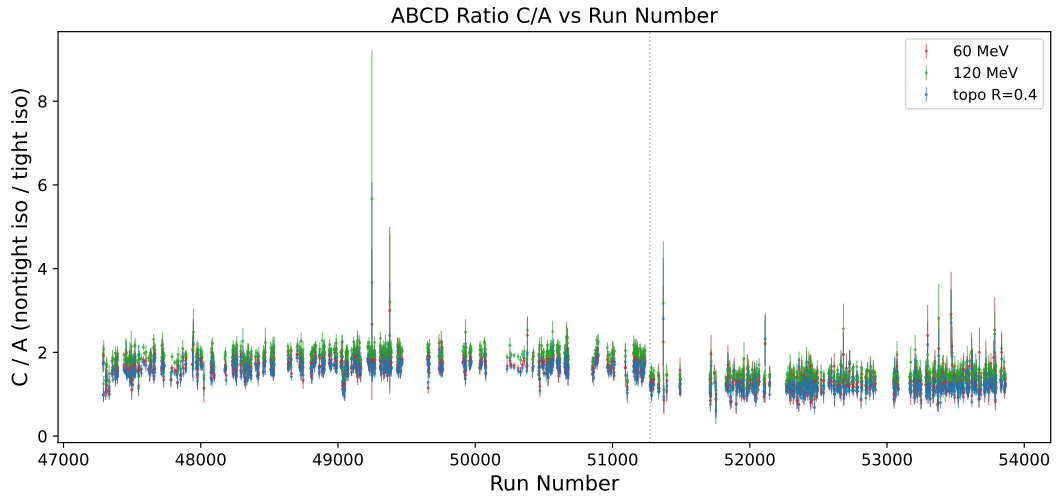


Figure 7: Ratio  $C/A$  (nontight-iso / tight-iso) vs. run number. All three definitions produce similar ratios, confirming that the ABCD closure is robust against the isolation definition choice.